

Coding & Solution

Date	29 June 2026
Team ID	xxxxxx
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the implementation quality, functionality, and completeness of the OptiCrop solution. The project successfully addresses the problem of crop selection by utilizing Machine Learning techniques to recommend the most suitable crop based on soil nutrients and environmental conditions. The solution follows clean coding practices, modular design principles, and a structured development approach to ensure maintainability, scalability, and ease of future enhancements.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/sasichintada/OptiCrop
Programming Language(s)	Python, HTML, CSS, JavaScript
Framework(s) Used	Flask, Bootstrap 5, Jinja2
Key Features Implemented	AI-Powered Crop Recommendation, Random Forest Prediction Model, Confidence Scores, Top Crop Matches, Data Preprocessing Pipeline, Exploratory Data Analysis, AI Assistant, Responsive User Interface, Model Evaluation and Visualization
Pending / Incomplete Features	Weather Forecast Integration, Fertilizer Recommendation System, Crop Yield Prediction, Mobile Application Support, IoT Sensor Integration

Setup / Run Instructions	1. Clone the repository. 2. Create and activate a virtual environment. 3. Install dependencies using requirements.txt.
	4. Run the application using python run.py. 5. Open http://127.0.0.1:5000 in a web browser.



Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments: